



Advanced Linear Algebra: Foundations to Frontiers

Robert van de Geijn, Margaret Myers

3.5.2 Summary ¶

Classical Gram-Schmidt orthogonalization: Given a set of linearly independent vectors $\{a_0, \dots, a_{n-1}\} \subset \mathbb{C}^m$, the Gram-Schmidt process computes an orthonormal basis $\{q_0, \dots, q_{n-1}\}$ that spans the same subspace as the original vectors, i.e.

$$\text{Span}(\{a_0, \dots, a_{n-1}\}) = \text{Span}(\{q_0, \dots, q_{n-1}\}).$$

The process proceeds as follows:

- Compute vector q_0 of unit length so that $\text{Span}(\{a_0\}) = \text{Span}(\{q_0\})$:

- $\rho_{0,0} = \|a_0\|_2$

- Computes the length of vector a_0 .

- $q_0 = a_0 / \rho_{0,0}$

- Sets q_0 to a unit vector in the direction of a_0 .

Notice that $a_0 = q_0 \rho_{0,0}$

- Compute vector q_1 of unit length so that $\text{Span}(\{a_0, a_1\}) = \text{Span}(\{q_0, q_1\})$:

- $\rho_{0,1} = q_0^H a_1$

- Computes $\rho_{0,1}$ so that $\rho_{0,1} q_0 = q_0^H a_1 q_0$ equals the component of a_1 in the direction of q_0 .

- $a_1^\perp = a_1 - \rho_{0,1} q_0$

- Computes the component of a_1 that is orthogonal to q_0 .

- $\rho_{1,1} = \|a_1^\perp\|_2$

- Computes the length of vector $a_1^\perp$.

- $q_1 = a_1^\perp / \rho_{1,1}$

- Sets q_1 to a unit vector in the direction of $a_1^\perp$.

Notice that

$$\left(\begin{array}{c|c} a_0 & a_1 \end{array} \right) = \left(\begin{array}{c|c} q_0 & q_1 \end{array} \right) \left(\begin{array}{c|c} \rho_{0,0} & \rho_{0,1} \\ \hline 0 & \rho_{1,1} \end{array} \right).$$

- Compute vector q_2 of unit length so that

$\text{Span}(\{a_0, a_1, a_2\}) = \text{Span}(\{q_0, q_1, q_2\})$:

- $\rho_{0,2} = q_0^H a_2$ or, equivalently, $\begin{pmatrix} \rho_{0,2} \\ \rho_{1,2} \end{pmatrix} = \begin{pmatrix} q_0 & q_1 \end{pmatrix}^H a_2$

- Computes $\rho_{0,2}$ so that $\rho_{0,2} q_0 = q_0^H a_2 q_0$ and $\rho_{1,2} q_1 = q_1^H a_2 q_1$ equal the components of a_2 in the directions of q_0 and q_1 .

- Or, equivalently, $\begin{pmatrix} q_0 & q_1 \end{pmatrix} \begin{pmatrix} \rho_{0,2} \\ \rho_{1,2} \end{pmatrix}$ is the component in $\text{Span}(\{q_0, q_1\})$.

- $a_2^\perp = a_2 - \rho_{0,2} q_0 - \rho_{1,2} q_1 = a_2 - \begin{pmatrix} q_0 & q_1 \end{pmatrix} \begin{pmatrix} \rho_{0,2} \\ \rho_{1,2} \end{pmatrix}$

- Computes the component of a_2 that is orthogonal to q_0 and q_1 .

- $\rho_{2,2} = \|a_2^\perp\|_2$

- Computes the length of vector $a_2^\perp$.

- $q_2 = a_2^\perp / \rho_{2,2}$

- Sets q_2 to a unit vector in the direction of $a_2^\perp$.

Notice that

$$\left(\begin{array}{c|c|c} a_0 & a_1 & a_2 \end{array} \right) = \left(\begin{array}{c|c|c} q_0 & q_1 & q_2 \end{array} \right) \left(\begin{array}{c|c|c} \rho_{0,0} & \rho_{0,1} & \rho_{0,2} \\ \hline 0 & \rho_{1,1} & \rho_{1,2} \\ \hline 0 & 0 & \rho_{2,2} \end{array} \right).$$

- And so forth.

Theorem 3.5.2.1. QR Decomposition Theorem. *Let $A \in \mathbb{C}^{m \times n}$ have linearly independent columns. Then there exists an orthonormal matrix Q and upper triangular matrix R such that $A = QR$, its QR decomposition. If the diagonal elements of R are taken to be real and positive, then the decomposition is unique.*

Projection a vector y onto the orthonormal columns of $Q \in \mathbb{C}^{m \times n}$:

$[y^\perp, r] = \text{Proj}^\perp_{Q_{\text{CGS}}}(Q, y)$ (used by CGS)	$[y^\perp, r] = \text{Proj}^\perp_{Q_{\text{MGS}}}(Q, y)$ (used by MGS)
$y^\perp = y$ for $i = 0, \dots, k-1$ $\rho_i := q_i^H y$ $y^\perp := y^\perp - \rho_i q_i$ endfor	$y^\perp = y$ for $i = 0, \dots, k-1$ $\rho_i := q_i^H y^\perp$ $y^\perp := y^\perp - \rho_i q_i$ endfor

Gram-Schmidt orthogonalization algorithms:

$[A, R] := \text{GS}(A)$ (overwrites A with Q)

$A \rightarrow \left(\begin{array}{c|c} A_L & A_R \end{array} \right), R \rightarrow \left(\begin{array}{c|c} R_{TL} & R_{TR} \\ \hline 0 & R_{BR} \end{array} \right)$
 A_L has 0 columns and R_{TL} is 0×0
while $n(A_L) < n(A)$

$\left(\begin{array}{c|c} A_L & A_R \end{array} \right) \rightarrow \left(\begin{array}{c|cc} A_0 & a_1 & A_2 \end{array} \right), \left(\begin{array}{c|c} R_{TL} & R_{TR} \\ \hline 0 & R_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c|cc} R_{00} & r_{01} & R_{02} \\ \hline 0 & \rho_{11} & r_{12}^T \\ \hline 0 & 0 & R_{22} \end{array} \right)$

CGS	MGS	MGS (alternative)
$r_{01} := A_0^H a_1$ $a_1 := a_1 - A_0 r_{01}$ $\rho_{11} := \ a_1\ _2$ $a_1 := a_1 / \rho_{11}$	$[a_1, r_{01}] = \text{Proj}_{\perp \text{to} Q_{\text{MGS}}}(A_0, a_1)$ $\rho_{11} := \ a_1\ _2$ $q_1 := a_1 / \rho_{11}$	$\rho_{11} := \ a_1\ _2$ $a_1 := a_1 / \rho_{11}$ $r_{12}^T := a_1^H A_2$ $A_2 := A_2 - a_1 r_{12}^T$

$\left(\begin{array}{c|c} A_L & A_R \end{array} \right) \leftarrow \left(\begin{array}{c|cc} A_0 & a_1 & A_2 \end{array} \right), \left(\begin{array}{c|c} R_{TL} & R_{TR} \\ \hline 0 & R_{BR} \end{array} \right) \leftarrow \left(\begin{array}{c|cc} R_{00} & r_{01} & R_{02} \\ \hline 0 & \rho_{11} & r_{12}^T \\ \hline 0 & 0 & R_{22} \end{array} \right)$

endwhile

Classic example that shows that the columns of Q , computed by MGS, are "more orthogonal" than those computed by CGS:

$$A = \left(\begin{array}{c|c|c} 1 & 1 & 1 \\ \hline \epsilon & 0 & 0 \\ 0 & \epsilon & 0 \\ 0 & 0 & \epsilon \end{array} \right) = \left(\begin{array}{c|c|c} a_0 & a_1 & a_2 \end{array} \right).$$

Cost of Gram-Schmidt algorithms: approximately $2mn^2$ flops.

Definition 3.5.2.2. Let $u \in \mathbb{C}^n$ be a vector of unit length ($\|u\|_2 = 1$). Then $H = I - 2uu^H$ is said to be a Householder transformation or (Householder) reflector.

If H is a Householder transformation (reflector), then

- $HH = I$.
- $H = H^H$.
- $H^H H = HH^H = I$.
- $H^{-1} = H^H = H$.

Computing a Householder transformation $I - 2uu^H$:

- Real case:
 - $v = x \mp \|x\|_2 e_0$.
 - $v = x + \text{sign}(\chi_1) \|x\|_2 e_0$ avoids catastrophic cancellation.
 - $u = v / \|v\|_2$
- Complex case:
 - $v = x \mp \boxed{\pm} \|x\|_2 e_0$.
 - (Picking $\boxed{\pm}$ carefully avoids catastrophic cancellation.)
 - $u = v / \|v\|_2$

Practical computation of u and τ so that $I - uu^H / \tau$ is a Householder transformation (reflector):

Algorithm : $\left[\begin{pmatrix} \rho \\ u_2 \end{pmatrix}, \tau \right] = \text{Housev} \left(\begin{pmatrix} \chi_1 \\ x_2 \end{pmatrix} \right)$	
$\rho = -\text{sign}(\chi_1) \ x\ _2$	$\chi_2 := \ x_2\ _2$
$\nu_1 = \chi_1 + \text{sign}(\chi_1) \ x\ _2$	$\alpha := \left\ \begin{pmatrix} \chi_1 \\ \chi_2 \end{pmatrix} \right\ _2 (= \ x\ _2)$
$u_2 = x_2 / \nu_1$	$\rho := -\text{sign}(\chi_1) \alpha$
	$\nu_1 := \chi_1 - \rho$
	$u_2 := x_2 / \nu_1$
	$\chi_2 = \chi_2 / \nu_1 (= \ u_2\ _2)$
$\tau = (1 + u_2^H u_2) / 2$	$\tau = (1 + \chi_2^2) / 2$

Householder QR factorization algorithm:

$[A, t] = \text{HQR_unb_var1}(A)$
$A \rightarrow \left(\begin{array}{c c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right)$ and $t \rightarrow \left(\begin{array}{c} t_T \\ t_B \end{array} \right)$ A_{TL} is 0×0 and t_T has 0 elements while $n(A_{BR}) > 0$ $\left(\begin{array}{c c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c c c} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right)$ and $\left(\begin{array}{c} t_T \\ t_B \end{array} \right) \rightarrow \left(\begin{array}{c} t_0 \\ \tau_1 \\ t_2 \end{array} \right)$ $\left[\begin{pmatrix} \alpha_{11} \\ a_{21} \end{pmatrix}, \tau_1 \right] := \left[\begin{pmatrix} \rho_{11} \\ u_{21} \end{pmatrix}, \tau_1 \right] = \text{Housev} \left(\begin{pmatrix} \alpha_{11} \\ a_{21} \end{pmatrix} \right)$ Update $\begin{pmatrix} a_{12}^T \\ A_{22} \end{pmatrix} := \left(I - \frac{1}{\tau_1} \begin{pmatrix} 1 \\ u_{21} \end{pmatrix} \begin{pmatrix} 1 & u_{21}^H \end{pmatrix} \right) \begin{pmatrix} a_{12}^T \\ A_{22} \end{pmatrix}$ via the steps $w_{12}^T := (a_{12}^T + a_{21}^H A_{22}) / \tau_1$ $\begin{pmatrix} a_{12}^T \\ A_{22} \end{pmatrix} := \begin{pmatrix} a_{12}^T - w_{12}^T \\ A_{22} - a_{21} w_{12}^T \end{pmatrix}$ $\left(\begin{array}{c c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \leftarrow \left(\begin{array}{c c c} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right)$ and $\left(\begin{array}{c} t_T \\ t_B \end{array} \right) \leftarrow \left(\begin{array}{c} t_0 \\ \tau_1 \\ t_2 \end{array} \right)$ endwhile

Cost: approximately $2mn^2 - \frac{2}{3}n^3$ flops.

Algorithm for forming Q given output of Householder QR factorization algorithm:

$[A] = \text{FormQ}(A, t)$
$A \rightarrow \left(\begin{array}{c c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right), t \rightarrow \left(\begin{array}{c} t_T \\ t_B \end{array} \right)$ A_{TL} is $n(A) \times n(A)$ and t_T has $n(A)$ elements while $n(A_{TL}) > 0$ $\left(\begin{array}{c c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c c c} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right), \left(\begin{array}{c} t_T \\ t_B \end{array} \right) \rightarrow \left(\begin{array}{c} t_0 \\ \tau_1 \\ t_2 \end{array} \right)$ Update $\begin{pmatrix} \alpha_{11} & a_{12}^T \\ a_{21} & A_{22} \end{pmatrix} :=$ $\left(I - \frac{1}{\tau_1} \begin{pmatrix} 1 \\ u_{21} \end{pmatrix} \begin{pmatrix} 1 & u_{21}^H \end{pmatrix} \right) \begin{pmatrix} 1 & 0 \\ 0 & A_{22} \end{pmatrix}$ via the steps $\alpha_{11} := 1 - 1/\tau_1$ $a_{12}^T := -(a_{21}^H A_{22}) / \tau_1$ $A_{22} := A_{22} + a_{21} a_{12}^T$ $a_{21} := -a_{21} / \tau_1$ $\left(\begin{array}{c c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \leftarrow \left(\begin{array}{c c c} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right), \left(\begin{array}{c} t_T \\ t_B \end{array} \right) \leftarrow \left(\begin{array}{c} t_0 \\ \tau_1 \\ t_2 \end{array} \right)$ endwhile

Cost: approximately $2mn^2 - \frac{2}{3}n^3$ flops.

Algorithm for applying Q^H given output of Householder QR factorization algorithm:

$[y] = \text{Apply_QH}(A, t, y)$
$A \rightarrow \left(\begin{array}{c c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right), t \rightarrow \left(\begin{array}{c} t_T \\ t_B \end{array} \right), y \rightarrow \left(\begin{array}{c} y_T \\ y_B \end{array} \right)$ A_{TL} is 0×0 and t_T, y_T have 0 elements while $n(A_{BR}) < 0$ $\left(\begin{array}{c c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c c c} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right),$ $\left(\begin{array}{c} t_T \\ t_B \end{array} \right) \rightarrow \left(\begin{array}{c} t_0 \\ \tau_1 \\ t_2 \end{array} \right), \left(\begin{array}{c} y_T \\ y_B \end{array} \right) \rightarrow \left(\begin{array}{c} y_0 \\ \psi_1 \\ y_2 \end{array} \right)$ Update $\begin{pmatrix} \psi_1 \\ y_2 \end{pmatrix} := \left(I - \frac{1}{\tau_1} \begin{pmatrix} 1 \\ u_{21} \end{pmatrix} \begin{pmatrix} 1 & u_{21}^H \end{pmatrix} \right) \begin{pmatrix} \psi_1 \\ y_2 \end{pmatrix}$ via the steps $\omega_1 := (\psi_1 + a_{21}^H y_2) / \tau_1$ $\begin{pmatrix} \psi_1 \\ y_2 \end{pmatrix} := \begin{pmatrix} \psi_1 - \omega_1 \\ y_2 - \omega_1 u_2 \end{pmatrix}$ $\left(\begin{array}{c c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \leftarrow \left(\begin{array}{c c c} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right),$ $\left(\begin{array}{c} t_T \\ t_B \end{array} \right) \leftarrow \left(\begin{array}{c} t_0 \\ \tau_1 \\ t_2 \end{array} \right), \left(\begin{array}{c} y_T \\ y_B \end{array} \right) \leftarrow \left(\begin{array}{c} y_0 \\ \psi_1 \\ y_2 \end{array} \right)$ endwhile

Cost: approximately $4mn - n^2$ flops.